

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

**LACTOSE TRYPTONE BROTH****LACTOSE TRYPTONE BROTH**

CM0931

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Details of the supplier of the safety data sheet**

Company

Thermo Scientific Microbiology Sdn Bhd  
No.6, Jalan TTC 6, Taman Teknologi Cheng,  
Cheng, 75250 Melaka, Malaysia  
+606 334 0975 .

Supplier

Oxoid Ltd.  
Wade Road  
Basingstoke, Hants, UK  
RG24 8PW  
Telephone: +44 (0) 1256 841144

E-mail address

mbd-sds@thermofisher.com

**Emergency Telephone Number**(603) 5122 8888  
CHEMTREC Malaysia **1-800-815-308** (Malay)  
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)**SECTION 2: HAZARDS IDENTIFICATION****Classification of the substance or mixture****Label Elements**

Signal Word

None

Hazard Statements

Precautionary Statements

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors

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## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                         | CAS No  | Weight % |
|-----------------------------------|---------|----------|
| Tris (hydroxymethyl) aminomethane | 77-86-1 | 1.1      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse thoroughly with plenty of water, also under the eyelids. Get medical attention if irritation persists.                                     |
| <b>Skin Contact</b>                       | Wash with plenty of soap and water. Get medical attention if irritation develops and persists.                                                   |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention if symptoms occur.                                                                                    |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

No information available.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

Use extinguishing method compatible with surroundings.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

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Avoid dust formation. Avoid contact with skin and eyes. Ensure adequate ventilation.

## Environmental precautions

Prevent further leakage or spillage if safe to do so.

## Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation. Do not breathe dust. Avoid contact with skin and eyes.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Protect from moisture.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### **Engineering Measures**

Provide appropriate exhaust ventilation at places where dust is formed.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

To protect the wearer, respiratory protective equipment must be the correct fit and be used

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and maintained properly  
When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No special environmental precautions required Avoid dust formation

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Appearance Light brown  
Physical State Powder  
Odor No information available  
Odor Threshold No data available  
pH 7.0 - 7.4 @ 25°C

Melting Point/Range Not applicable  
Softening Point No data available  
Boiling Point/Range Not applicable  
Flash Point Not applicable

Method - No information available

Evaporation Rate No data available  
Flammability (solid,gas) No information available  
Explosion Limits No data available

Vapor Pressure No data available  
Vapor Density No data available (Air = 1.0)  
Specific Gravity / Density No data available  
Bulk Density No data available  
Water Solubility No information available  
Solubility in other solvents No information available

### Partition Coefficient (n-octanol/water)

Autoignition Temperature Not applicable  
Decomposition Temperature No data available  
Viscosity No data available  
Explosive Properties No information available  
Oxidizing Properties No information available

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

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Stable under recommended storage conditions. Hygroscopic.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Protect from direct sunlight. Protect from moisture. Avoid dust formation.

## Incompatible Materials

Strong oxidizing agents.

## Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Acute Toxicity

| Component                         | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation |
|-----------------------------------|---------------------------|---------------------------|-----------------|
| Tris (hydroxymethyl) aminomethane | LD50 = 5900 mg/kg ( Rat ) | LD50 > 5000 mg/kg ( Rat ) |                 |

#### Chronic Toxicity

##### **Carcinogenicity**

There are no known carcinogenic chemicals in this product

##### **Sensitization**

None known

##### **Mutagenic Effects**

None known

##### **Reproductive Effects**

None known

##### **Developmental Effects**

None known

##### **Target Organs**

No information available.

##### **Neurological Effects**

None known

#### **Endocrine Disruptor Information**

None known

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

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|                                      |                                                                |
|--------------------------------------|----------------------------------------------------------------|
| <u>Persistence and degradability</u> | Expected to be biodegradable                                   |
| <u>Bioaccumulative potential</u>     | Bioaccumulation is unlikely                                    |
| <u>Mobility in soil</u>              | The product is water soluble, and may spread in water systems. |
| <u>Other adverse effects</u>         | None known                                                     |

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products** Dispose of in accordance with local regulations

**Contaminated Packaging** Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <u>IMDG/IMO</u>                     | Not regulated                   |
| <u>Road and Rail Transport</u>      | Not regulated                   |
| <u>IATA</u>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component                         | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------------|--------|------|-----|-------|------|------|-------|------|----------|
| Tris (hydroxymethyl) aminomethane | -      | X    | X   | X     | X    | X    | X     | X    | KE-01403 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Prepared By

Revision Date

Revision Summary

Regulatory Affairs

28-Mar-2023

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**